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Metric Spaces

1.1 Convergence of Sequences

Definition 1.1.1 ► Convergence

Let (X, d) be a metric space and $x_1, \dots, x_n, \dots$ a sequence of points in X . Then the sequence is said to **converge to** $x \in X$ if given any $\varepsilon > 0$ there exists an integer n_0 such that for all $n \geq n_0$, $d(x, x_n) < \varepsilon$. This is denoted by $x_n \rightarrow x$.

The sequence $y_1, y_2, \dots, y_n, \dots$ of points in (X, d) is said to be **convergent** if there exists a point $y \in X$ such that $y_n \rightarrow y$.

Proposition 1.1.2

Let (X, d) be a metric space. A subset A of X is closed in (X, d) if and only if every convergent sequence of points in A converges to a point in A . (In other words, A is closed in (X, d) if and only if $a_n \rightarrow x$, where $x \in X$ and $a_n \in A$ for all n , implies $x \in A$.)

1.2 Completeness

Definition 1.2.1 ► Cauchy sequence

A sequence $x_1, x_2, \dots, x_n, \dots$ of points in a metric space (X, d) is said to be a **Cauchy sequence** if given any real number $\varepsilon > 0$, there exists a positive integer n_0 , such that for all integers $m \geq n_0$ and $n \geq n_0$, $d(x_m, x_n) < \varepsilon$.

Proposition 1.2.2

Let (X, d) be a metric space and $x_1, x_2, \dots, x_n, \dots$ a sequence of points in (X, d) . If there exists a point $a \in X$, such that the sequence converges to a , that is, $x_n \rightarrow a$, then the sequence is a Cauchy sequence.